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B-78

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Array: [45, 22, 67, 19, 50, 89, 33, 72, 40, 55]

Q1. Construct a binary search tree by inserting the given array elements sequentially. Then write the in order traversal.

Q2. Convert the given array into a Max-Heap using the bottom up process, show the resulting array after each major heapify step. Also state the time complexity of heapify

Q3. Using the given array, insert them keys into a hash table of size 10 using hash function h(key) = key % 10. Resolve collision using linear probing. Show the final hash table. What is the load factor after all the insertions.

Q4. Insert the first 6 elements into an initially empty AVL tree. Identify the violation point, and show the rotation performed. Also state the balance factor after all insertions.

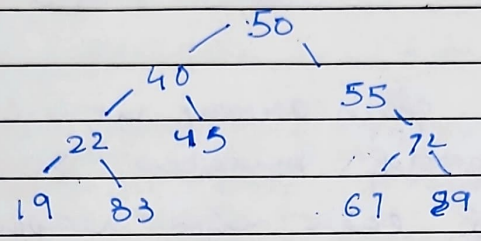
Q5. Treat the given array as list of vertices. Create an undirected graph by connecting each adjacent pair (45-22, 22-67, 67-19, 19-50, 50-89, 89-33, 33-72, 72-40, 40-55) into array. Represent this graph using
a) Adjacency Matrix
b) Adjacency list
Also mention the space complexity of each representation.

[45, 22, 67, 19, 50, 89, 33, 72, 40, 55]
(19, 22, 33, 40, 45, 50, 55, 67, 72, 89)

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Q1

1. Adding sequentially:
since the smallest element is 19
it will be in the last node



(19 22 33 40 45 50 55 67 72 89)

Step by step adding:

Add → 50 → 50

Add 40 → 50
40

Add 22 → 50
40
22

Add 19 → 50
40
22
19

Add 33 → 50
40
22
19 33

Add 45 → 50
40
22 45
19 33

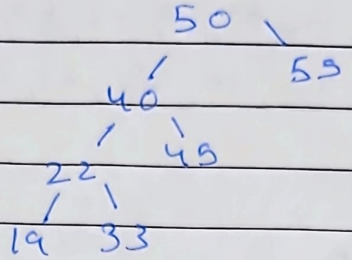
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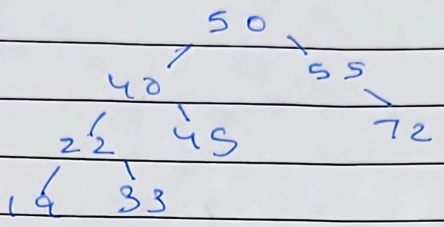
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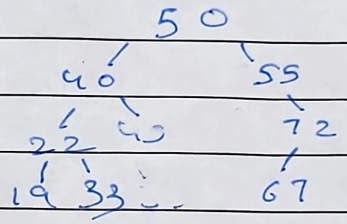
Add 55 →



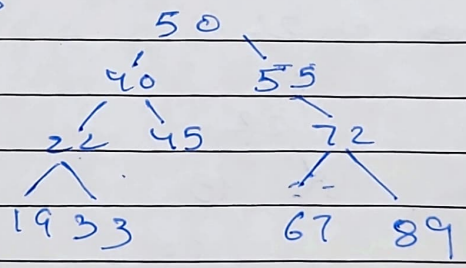
Add 72 →



Add 67 →



Add 89 →



Inorder: [19, 22, 33, 40, 45, 50, 55, 67, 72, 89]

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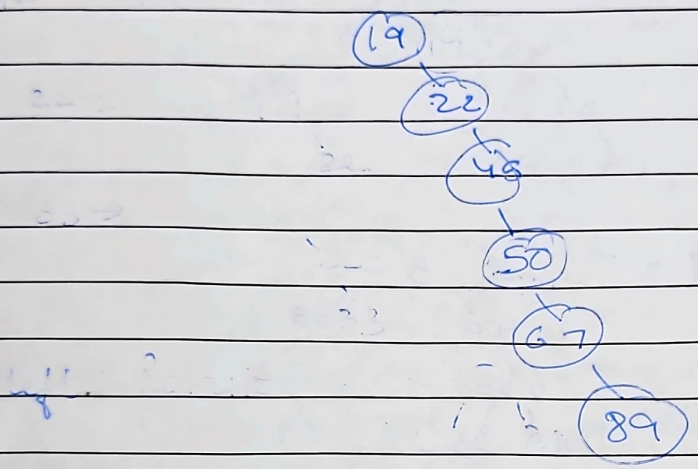
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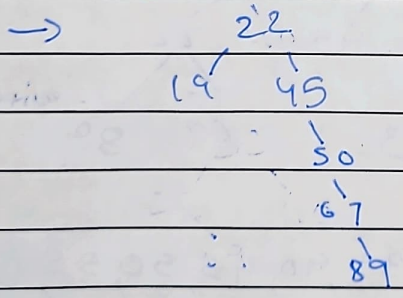
Q4 [45, 22, 67, 19, 50, 89, 33, 72, 40, 55]

[19, 22, 33, 40, 45, 50, 55, 67, 72, 89]

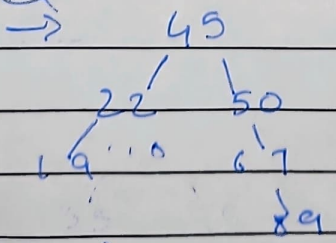
Adding Elements



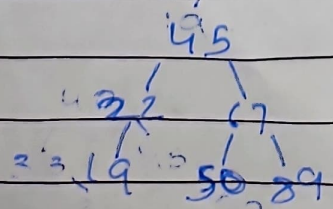
Left rotation on 22



Left rotate on (45)



Left rotate 67



Quick Work

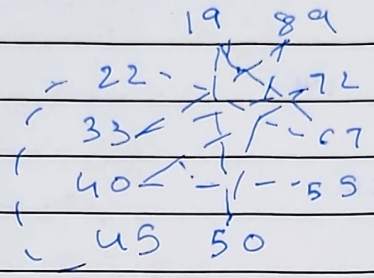
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Balance factor = Left H − Right H — ①

Q5 45-22, 22-67, 67-19, 19-50, 50-89, 89-33, 33-72
72-40, 40-55



Matrix

	19	22	33	40	45	50	55	67	72	89
19	0	0	0	0	0	1	0	1	0	0
22	0	0	0	0	1	0	0	1	0	0
33	0	0	0	0	0	0	0	0	1	1
40	0	0	0	0	0	0	1	0	1	0
45	0	1	0	0	0	0	0	0	0	0
50	1	0	0	0	0	0	0	0	0	1
55	0	0	0	1	0	0	0	0	0	0
67	1	1	0	0	0	0	0	0	0	0
72	0	0	1	1	0	0	0	0	0	0
89	0	0	1	0	0	1	0	0	0	0

Adj List :

19 (50, 67)	67 (19, 22)
22 (45, 67)	72 (33, 40)
33 (72, 89)	89 (33, 50)
40 (55, 72)	
45 (22)	
50 (19, 89)	
55 (40)	

Space Complexity of Matrix
$= O(10 \times 10)$

Space Complexity of list
$= O(10n)$

Q2

Heap / - 19

22 — 22

19

33 — 22

1

Q3 19 → 19

22 →

20

19

40 → 40

19 22

72

72 → 40

19 22

55 → 72

40

19 22

55 →